Supraharmonics: Concepts and Experimental Results on Photovoltaic Systems

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***Abstract* — The characterization of supraharmonics referring to conducted disturbances at frequencies above 2 kHz is gaining interest because of their interaction with the range of frequencies occurring in practical applications such as smart meter power-line communication. This paper reviews various aspects concerning the measurement and representation of the supraharmonics, as well as approximated modelling that can be used in the presence of inverters. Experimental results referring to photovoltaic systems are included.**

***Keywords* — supraharmonics, power quality, photovoltaic, meters, grid connection, measurement.**

I. INTRODUCTION

For harmonic analysis, the classical limits to the harmonic orders set by the Standards and implemented in the power quality analyzers are the 40th (or the 50th) harmonic. For a 50 Hz system, the 40th and 50th harmonics mean 2 kHz and 2.5 kHz, respectively, and the corresponding frequencies for a 60-Hz system are 2.4 kHz and 3 kHz. In practice, under this view all the conducted disturbances exceeding these frequencies are not taken into account in the determination of the classical harmonic distortion indicators implemented in the analyzers.

However, the number of applications in which non-negligible conducted disturbances appear at frequencies above 2 kHz has increased during the years, also because no strict regulation has been in place concerning disturbances at relatively high frequencies. In the last decade there has been a growing interest towards the so-called *supraharmonics* (or high-frequency harmonics), i.e., harmonics appearing at frequencies above 2 kHz. The related aspects are fully considered among the current issues for harmonic analysis [1][2].

In early studies (e.g., [3]) the supraharmonics have been addressed in the frequency range from 2 to 9 kHz. Nowadays, the frequency range can be extended up to 100 kHz or 150 kHz, because the power-line communications used in smart meters work in this spectrum zone, and the interaction between power-line communications and different types of end-user equipment may affect the equipment performance [4]. Therefore, it is possible a negative interaction between the smart-meter signals and the supraharmonics produced by grid-connected harmonic sources, both on the demand side and on the generation side, e.g., from photovoltaic (PV) inverters.

Supraharmonics are a timely topic being addressed by a number of Working Groups in all the Committees and associations relevant to the evolution of the power quality Standards [5]. A recent contribution [6] addresses the issue of the standardization for supraharmonics. In particular, the existing standards that can be applied in the frequency range 2 to 150 kHz are described and compared.

Further documents have been prepared by different standardization bodies. In particular, a convergence of interests is emerging and is reflected in the upgrade of the power quality standards accessing supraharmonics frequency levels from the "low frequency" side (below 150 kHz)[7] and from the "high frequency" side (above 150 kHz). The latter are interested in extending the frequency range to lower levels, in order to ensure appropriate EMC performance of the equipment connected in the smart grid. An example is the International standardization activity of the International Special Committee on Radio Interference (CISPR), belonging to the International Electrotechnical Commission (IEC), currently working towards establishing new requirements in the band below 150 kHz, also pointing out to which equipment the new limits should be applied [8].

Various types of equipment have been studied with the aim of highlighting the supraharmonics appearing during their operation. Main subjects of investigation have been fluorescent lights [9], high-frequency ballasts (with the relevant effects on their lifetime) [10] and LED lamps [11]. Other examples include residential equipment such as TV sets and microwave ovens [12], with further addition of the effect of electric vehicles in [13]. Experimental results referring to groups of loads are provided in [14]. Propagation of high-frequency harmonics in industrial networks has been addressed in [15], taking into account conventional loads and variable speed drives.

The assessment of supraharmonics referring to distributed generation with wind and PV systems is particularly challenging, because of the uncertain behavior of the sources depending on ambient conditions, the presence of conversion systems and the effect of harmonic summation for aggregations of local generators, also taking into account the connecting circuits. Wind systems have been addressed in [16], pointing out the attention to individual wind turbines and especially on the effects of their aggregation in wind farms. An emergent need to refine the models and study the propagation of the harmonic emission to the grid is indicated. Further analyses on wind parks at frequencies up to 200 kHz have been reported in [17] by using the concept of overall transfer function, concluding that in the case studied the

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emission of harmonics at frequencies above a few kilohertz has been negligible. A number of contributions referring to PV systems are reviewed in Section II of this paper.

One of the open questions is the characterization of the supraharmonics for an aggregation of sources. Initial results are borrowed from harmonic analysis, considering that, when the frequencies are relatively high, the phase angle dispersion becomes relevant. In case of randomly selected phase angles for a number of aggregated components of equal characteristics, the amplitude of the aggregated harmonic is proportional to the inverse of the square root of the number of components [18].

This paper recalls some emergent concepts referring to the presence of supraharmonics in electrical distribution system applications, including distributed generation, and presents some inputs concerning the experimental characterization of supraharmonics in PV systems.

The next sections are organized as follows. Section II summarizes some literature contributions referring to harmonics and supraharmonics in PV systems. Section III addresses measurement aspects and the representation of the results. Section IV deals with modeling aspects for the interpretation of the supraharmonics contents. Section V presents the results of experimental tests carried out on a real PV system. The last section contains the concluding remarks.

II. Harmonics and Supraharmonics in Photovoltaic Systems – An Overview

Many references can be found in the literature with respect to the emissions caused by photovoltaic generators connected to LV and MV networks up to 2 kHz. For instance, [19] gives an overview of the current state of knowledge within the context of IEC regulations, particularly in regards to harmonics, flicker, and voltage unbalance.

Experimental data based on PV system measurement data to verify compliance with the PQ requirements are presented in [20]. Results of further investigations, also considering the interaction of several inverters connected at the same busbar, can be found in [21]-[23]. In [24] the harmonic current emission in PV systems is characterised under real operating conditions, and a statistical methodology for assessing PV harmonic current emission is proposed.

Some recent contributions focused on the emission characterization at frequencies greater than 2 kHz. Few papers report the results of measurement campaigns on PV installations focused on the analysis of harmonic emission both for classical harmonics below 2 kHz and higher frequency emission in the range of 2-150 kHz. In particular, [25] reports the results of extensive measurements carried out at a 1 MW PV installation with multiple inverters with different rated power, at frequencies up to 100 kHz. High frequency current emissions are analysed for a single inverter and for a group of inverters. From the results reported, it is also possible to observe the effect of narrow band power-line communication systems (PLC), whose contributions from different channels appear as multiple increases in the current spectrum (one for each channel) in the frequency range from 40 kHz to 80 kHz. The presence of multiple channels is useful to enable switching from one channel to another in case of disturbance to the communications.

Reference [26] focused on PV installations connected to LV grids, where PV generators generally use inverters that generate the output current using pulse width modulation (PWM). Compared with other techniques, this efficiently reduces low order harmonics, but leads to high frequency components around the PWM carrier signal frequency (typically between 15 kHz and 20 kHz) and its multiples. Different frequency bands are implemented and compared. Differences between the distribution of the harmonics in the three phases for PV plants connected to the three-phase system are also shown. Peaks in supraharmonics emissions have been reported in [27]. In particular, a peak around 16 kHz is likely to be associated with the switching of the inverter. Confirmation of this fact also comes from the absence of this peak during the night, without PV production.

For PV systems, the PWM technology, commercially used in single-phase and three-phase connections, exhibits unipolar and bipolar configurations with triangle waveforms in the range 1-20 kHz. The harmonic emissions are different according to the type of configuration, however the harmonics are grouped into integer multiples of the triangular carrier [28]. In particular, for relatively small inverters (below 100 kW) the switching frequency is generally above 10 kHz, while for larger inverters the switching frequency is generally below 10 kHz [1].

Experimental results from [29] showed the presence of 3.8 kHz and 4.85 kHz components, poorly correlated with the PV production. These components are associated with the power-line communications exploited by the distribution system operator to transmit data referring to automated meter readings. Some conclusions from [30] are that every installation is unique, and for large numbers of inverters the level of distortion in the frequency range from 2 kHz to 150 kHz requires further study, with particular respect to aggregation effects, the spread of emission, the interaction between different typologies of devices, and the relation with active power production.

III. Measurements and Representation of the Results

One of the major issues in the research on supraharmonics regards the measurement methods that can be applied when measuring signal of voltage and current for frequency greater than 2 kHz.

One of the earlier contributions with respect to high-frequency distortion was the examination of the results from a measurement campaign in the frequency range 2 - 150 kHz [31]. In [31] time-domain, frequency-domain and a combined time-frequency domain representations of the waveform are used. Measurements of both current emission and voltage distortion are presented.

Concerning the comparison between some ways to measure the harmonics and supraharmonics components, the standards for harmonics measurement (IEC 61000-4-7 and IEC 61000-4-30) identify a 10-cycle time window (200 ms for a 50-Hz system) over which time-domain data are sampled. Then, the signal is processed through Discrete Fourier Transform (DFT) and the results are arranged into frequency bands of 5 Hz width. Harmonic groups and subgroups are then defined. Reference [3]

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discusses the implications of considering 200 ms window (leading to 5 Hz components) for frequencies below 2 kHz and 100 ms window (leading to 10 Hz components) from 2 to 9 kHz, arranging the results in 200 Hz frequency bands. A safe limit is indicated to be 0.5% of nominal voltage per 200-Hz band.

The issue of measurement methods for voltage and current in the frequency range of 2 kHz to 150 kHz is addressed in [32]. In particular, two measurement methods are compared: the method for frequencies of 2 kHz to 9 kHz proposed in the informative annex of IEC 61000-4-7, which could be extended in application to 150 kHz, and the method for the frequency range of 9 kHz to 150 kHz proposed by the committee draft for vote of IEC 61000-4-30 Ed.3.

An oscilloscope having only 8 bits is not sufficient [15]. A 16 bit resolution system is needed, e.g., with maximum sampling frequency of 250 kSa/s. In this way, the frequency limit depends on the number of channels used. From the Nyquist-Shannon theorem, for a single channel the maximum frequency is of 125 kHz, and is halved (to 62.5 kHz) if two channels are used.

The frequency bandwidth of the probes is also a primary concern. Significant differences in the results obtained by using different probes in the same system have been reported in [15]. There are different instruments to measure AC current signals: the most used categories are the Hall-effect current clamps and the Rogowski coils. The accuracy of the Hall-effect probes is higher than the other sensors. However, even if the first category permits to extend the frequency range down to the zero frequency, the second category permits to increase the frequency band up to some tens of kilohertz.

In the experimental campaign reported in this paper, the frequency ranges are from DC to 15 kHz for the Hall-effect current probes with peak values of 2000 A, while the same ranges are from 10 Hz to 50 kHz for the Rogowski coils. A correct positioning of the conductors inside the probe cores is crucial for both the types of instruments: normally, the available space to embrace the conductor is higher in the case of Rogowski coil. This sensor is made of a flexible coil of wire, without iron core, and it is wrapped around the conductor whose current must be measured. Other factors which affect the measurements are the winding density, the coil diameter and the rigidity of the winding. In the Rogowski coil, the induced voltage is proportional to the derivative of the current in the conductor under study. Hence, the output of the Rogowski coil is connected to an integrator circuit to provide an output signal proportional to the current.

For representation purposes, time-domain graphs and harmonic spectra are generally used. Due to the wide range of harmonics considered and to the specific interest towards the supraharmonics, logarithmic representations are also used (e.g., with the amplitude scale expressed in decibels; typical units for voltage and current amplitudes are $dB_{\mu V}$ and $dB_{\mu A}$, respectively [25]).

About the techniques of analysis, the classical time-domain and frequency-domain analysis are used, while some contributions in identifying the supraharmonics signatures of different systems may come from the results of time-frequency analysis [14].

IV. Modelling Aspects

Harmonic and interharmonic penetration studies require accurate models of power system components (lines, cables, transformers, electrical machines and loads) in the frequency range of interest. They are historically based on the study of the frequency range 0 - 2.5 kHz. Some attention has been devoted to the frequency range 2.5 - 9 kHz, which is the upper limit of the low-frequency range considered by IEC Standards. Recently, starting from the consideration that modern converters utilize high commutation frequencies, the attention has been extended to the frequencies up to 100 kHz. In [33] contributions for modelling behaviour of both overhead and cable MV lines in the frequency range from 0 to 100 kHz are given.

The spread of high-frequency current between different devices and into the grid is explained in [4] by means of a model also used to show the aggregation between devices and to predict the amplitude modulation of the voltage and current ripple due to the small frequency differences between devices.

The understanding of the real situation, when a PV inverter (or a group of PV inverters) is connected to a low voltage grid, is possible by using a simple equivalent circuit, in which both the Voltage Source Inverter (VSI) and the distribution grid are represented by the corresponding Thévenin equivalents. The equivalent circuit of each device in an electric power system is an appropriate tool to simulate the behavior of this system in terms of harmonic components. In particular, the Thévenin generator is the most used equivalent circuit for both the power supplies (e.g. the distributed generators and the distribution grid) [15]. Regarding the loads connected to the grid, besides the usual constant impedance model, it is considered a current source, with the proper harmonic components, to simulate the current injection into the grid, according to the frequency under study in the operation of the electric power system [2].

The parameters are defined for each harmonic order *h* of the frequency spectrum in this way: ideal voltage sources $\overline{V}_{vsi}^{h}$, $\overline{V}_{grid}^{h}$ and equivalent impedances $\overline{Z}_{vsi}^{h}$ and $\overline{Z}_{grid}^{h}$. However, the harmonic components of the ideal voltage source $\overline{V}_{grid}^{h}$ are typically within the range 50-2500 Hz (excluding possible signals coming from the communication systems used in smart meters). Therefore, at the Point of Common Coupling (PCC) the current injected into the grid $\overline{I}_{pcc}^{h}$ and its voltage $\overline{V}_{pcc}^{h}$ can be calculated by using the common mesh equation and the Millman's theorem.

The quantitative analysis performed thanks to these equations shows that, neglecting the grid impedance with respect to the VSI impedance, both current and voltage have a spectrum similar to the spectrum generated by the PV inverter. Obviously, the PV inverter is equipped with the EMI filter to reduce the harmonic emission and thus the PV inverter spectrum is not identical to an ideal PWM spectrum.

In summary, starting from the basic equations and neglecting the effect of the load:

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$$\bar{I}_{pcc} = \frac{\bar{V}_{vsi} - \bar{V}_{grid}}{\bar{Z}_{vsi} + \bar{Z}_{grid}} \rightarrow \bar{I}^h_{pcc} = \frac{\bar{V}^h_{vsi} - \bar{V}^h_{grid}}{\bar{Z}^h_{vsi} + \bar{Z}^h_{grid}} \quad (1)$$

$$\bar{V}_{pcc} = \frac{\frac{\bar{V}_{vsi}}{\bar{Z}_{vsi}} + \frac{\bar{V}_{grid}}{\bar{Z}_{grid}}}{\frac{1}{\bar{Z}_{vsi}} + \frac{1}{\bar{Z}_{grid}}} \rightarrow \bar{V}^h_{pcc} = \frac{\frac{\bar{V}^h_{vsi}}{\bar{Z}^h_{vsi}} + \frac{\bar{V}^h_{grid}}{\bar{Z}^h_{grid}}}{\frac{1}{\bar{Z}^h_{vsi}} + \frac{1}{\bar{Z}^h_{grid}}} \quad (2)$$

when supraharmonics are considered, the equations can be approximated as:

$$\bar{I}^h_{pcc} \approx \frac{\bar{V}^h_{vsi}}{R_{vsi} + jh\omega L_{vsi}} \quad (3)$$

$$\bar{V}^h_{pcc} \approx \frac{R_{grid} + jh\omega L_{grid}}{R_{vsi} + jh\omega L_{vsi}} \cdot \bar{V}^h_{vsi} \quad (4)$$

Critical aspects of using these model equations are that the parameters are not constant in time and are difficult to be characterized. Approximated models may be useful to explain the outcomes occurring in specific situations.

V. Experimental Tests and Results

Experimental tests have been performed to characterize the harmonic content of the current and voltage at PCC for frequencies higher than 2 kHz.

A real PV system (850 kW_p) is considered [34]. The PV system is installed on the top of an industrial building in the outskirts of a big city. It is used partly to supply the local load, partly for power injection into the grid. Fig. 1 shows the layout of the PV system.

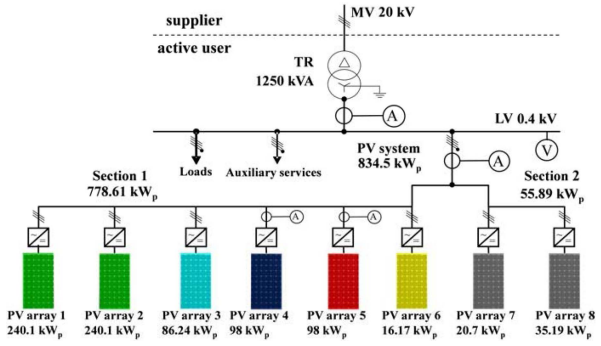


Fig. 1. Layout of the system with PV arrays and local loads.

The whole PV generator is composed of two different subfields of 800 kW_p (integrated into the shed roof) and 50 kW_p (non-building integrated) with mono-crystalline and poly-crystalline silicon modules, respectively. All the PV arrays are connected on the LV side through eight three-phase inverters with different power ratings: six for the first section and two for the second one. The inverters and the LV/MV transformer are oversized to minimize losses and maximize lifespan.

The total rated powers are 870 kVA for the inverters and 1250 kVA for the delta-star transformer. The PV arrays have been installed with the aim to maximize the occupation of the roof surface on the commercial building, accepting the occurrence of partial shading during the day on some arrays. The first field is oriented to the southwest; consequently, the exposure to the solar radiation is neither symmetrical nor optimal and is different during the morning and the evening. The second field is optimally (South) oriented and not integrated (less thermal losses). Two PV arrays (numbers 4 and 5) are subject to systematic partial shading.

Concerning the measurements, time-domain sampling of the voltage and current waveforms has been carried out by means of a portable data logging system. In order to reproduce the outputs within a wide frequency band, the instrumentation must have suitable accuracy. In this regard, in low voltage systems the voltages can be measured with percent uncertainty of about ±0.1% for frequencies up to some megahertz. However, the bottleneck is the measurement of the currents. Indeed, when a current clamp meter is used for peak current ranges of ±200/2000 A, the accuracy is around ±1.5% and a limit of 15 kHz is typical. The frequency limit can be increased if the currents are lower, for example with a peak current of ±30 A the frequency range can be extended up to 100 kHz [15].

The experimental results are ordered with a top-down approach. Voltage and current signals have been gathered in the three-phase four-wire system at different locations, starting from the low voltage side of the distribution transformer, continuing with the PV system (i.e., excluding loads and auxiliary services) and finishing with the inverter supplied by the PV array #4, subject to partial shading in a part of the day.

At the distribution transformer busbars the harmonic content of the local load supplied by the grid can be separated from the harmonic content of the PV system. For these measurements, the Rogowski coil has been used. Thanks to a dedicated test in which the general circuit breaker of the PV system is opened, the signals before and after this commutation are gathered. Fig. 2 shows the currents in the three phases before and after opening the general circuit breaker of the PV system.

Fig. 3 shows that the harmonic current of the PV system together with the load is characterized by peaks with groups of harmonic orders at multiples of 3 kHz. These harmonics are associated with the PV inverter operation. In order to verify this statement, by opening the circuit breaker and leaving only the supply to loads and auxiliary services, these peaks disappear (Fig. 4).

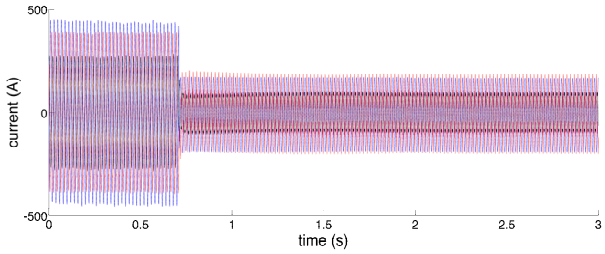


Fig. 2. Phase currents before and after opening the general circuit breaker of the PV system.

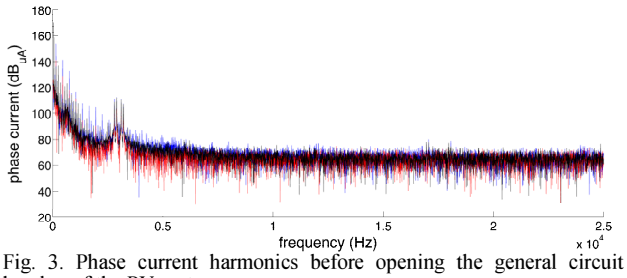


Fig. 3. Phase current harmonics before opening the general circuit breaker of the PV system.

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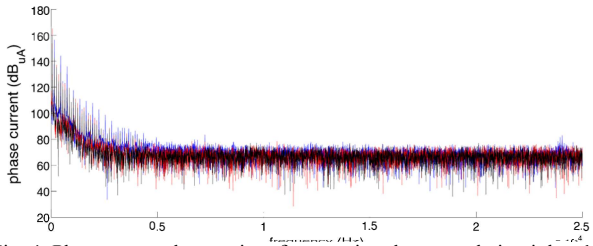


Fig. 4. Phase current harmonics after opening the general circuit breaker of the PV system (load and auxiliary services only).

Further measurement has been carried out for the PV system alone, again using the Rogowski coil. Fig. 5 shows the phase current harmonics in this case, highlighting the frequencies up to 10 kHz. The presence of the abovementioned peaks for groups of harmonic orders around multiples of 3 kHz is confirmed. This behavior is typical of inverters with power rating above some tens of kilowatts. From Fig. 5 it can also be observed that the phase currents are significantly unbalanced. The zoom reported in Fig. 6 shows that in the three phases in unbalanced conditions the peaks behave in a similar way.

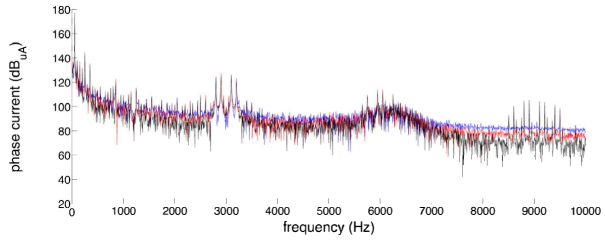


Fig. 5. Phase current harmonics (PV system only).

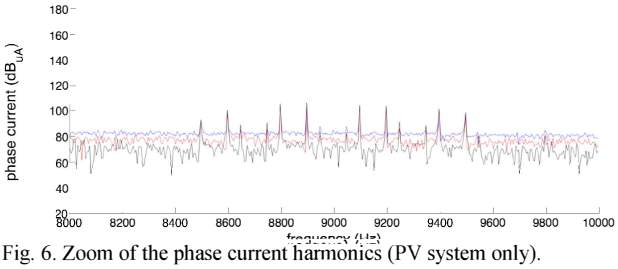


Fig. 6. Zoom of the phase current harmonics (PV system only).

Finally, at the level of the inverter supplied by PV array #4 subject to partial shading, the measurements have been carried out by using a current clamp, with limited frequency band. The results are presented in Fig. 7, up to 10 kHz. The previous behavior around multiples of 3 kHz can be found again. The mismatch in the *I-V* curves of the array [34] worsens the harmonic content of the currents at the AC output of the three-phase inverter, especially for low harmonic orders.

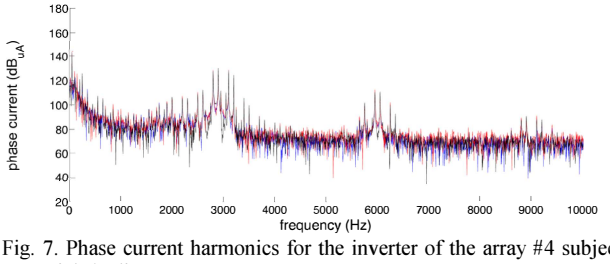


Fig. 7. Phase current harmonics for the inverter of the array #4 subject to partial shading.

In order to highlight possible limits in the execution of the measurements, the two graphs of Fig. 8 show the harmonic spectrum referring to the current of one of the phases in the measurement of the whole PV system. The amplitude at fundamental frequency (about 700 A) is truncated. The graph on the top contains the harmonic spectrum used in the classical harmonic analysis, up to 2 kHz. The graph on the bottom contains the harmonic spectrum relevant to the study on the supraharmonics, starting from 2 kHz and continuing up to 20 kHz. The measurement carried out with the Rogowski coil makes it possible to represent the correct information in the whole frequency range. If the measurement are done by using a current clamp, the data shown in the encircled part of the spectrum (above 15 kHz) are not represented, as for the system operating with relatively high currents (hundreds of amperes) the frequency range to be considered for the current clamp is limited to 15 kHz.

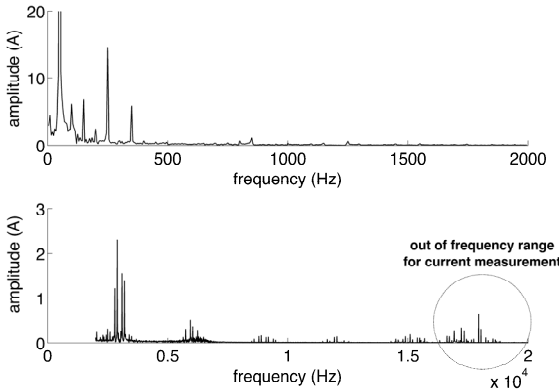


Fig. 8. Harmonic amplitude spectra of one phase current of the grid-connected PV system.

VI. Conclusion

The presence of supraharmonics in the electrical distribution systems calls for extending the present power quality rules and standards. This paper has provided an overview of the relevant concepts, based on specific literature contributions, addressing in particular the limitations in the frequency range of the current probes used to carry out measurement campaigns on large PV systems. Some results of experiments carried out by the authors on a large PV system have been reported in an appropriate scale based on decibels to make the main components of the harmonic currents clearly visible and to highlight the effects of current unbalance in the three-phase system. These results are useful to continue the discussion concerning the upgrade of power quality standards and grid codes, in particular those used in relatively weak distribution systems (such as the ones existing in islands) in which the impact of the local generation systems, for example from renewable sources, is not negligible with respect to the contribution of the main grid [35]. Specific attention also refers to the interactions of supraharmonics with power line communications, in particular concerning the five channels established by the CENELEC 50065.1 standard in the frequency band from 3 kHz to 148.5 kHz [36], of which the band limited to 95 kHz is dedicated to the electricity suppliers and their licensees, while the channels in the remaining frequency band are assigned to the customers.

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